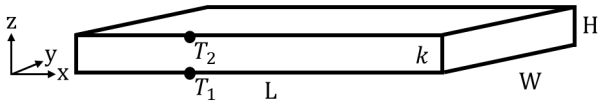
CBE 150B F2026
Problem Set 1

Assigned: 8/26/2026
Due: 9/2/2026 at 12:00 pm

Problem 1 (10 pts)

You are testing a novel thermal interface material (TIM) long, wide, thin film for microelectronics applications by imposing a temperature gradient across the thickness of the film. You hold the bottom surface at T_1 and the top surface at temperature T_2.

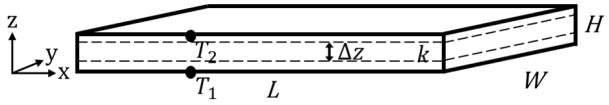


a) If $T_2 > T_1$, in which direction is heat transported? What is causing the heat transport? (2 pts)

Heat is transported in the negative z direction by conduction. Microscopically, this is driven by random collisions between particles with different average kinetic energies.
(+1 pt: heat directed downward from hot to cold, +1 pt: Difference in kinetic energy of particles at top and bottom of the slab.)

b) Starting from the generalized energy balance (written in words then translated into an equation), derive the steady state temperature profile $T(z)$ in the slab. (4 pts)

in-out+generation-consumption=accumulation (+1 pt)
in=out (+1 pts)
If solving via shell balance, here's a quick reminder about what that looks like:



$q_z|_z \cdot LW = q_z|_{z+\Delta z} \cdot LW$
Divide by volume $(\div \Delta z \cdot LW)$ $\frac{q_z|_z}{\Delta z} = \frac{q_z|_{z+\Delta z}}{\Delta z}$
$\lim_{\Delta z \to 0} \frac{q_z|_{z+\Delta z} - q_z|_z}{\Delta z} = 0$ (+2 pts)
$\frac{\partial q_z}{\partial z} = 0$
Plug in Fourier's Law $(q_z = -k\frac{\partial T}{\partial z}) \Rightarrow 0 = -k\frac{\partial^2 T}{\partial z^2}$ (+2 pts)
Integrate twice:
$T(z) = C_1 T + C_2$
Plug in BCs: (i) $T(z=0) = T_1$, (ii) $T(z=H) = T_2$

$$\boxed{T(z) = \frac{T_2 - T_1}{H} z + T_1}$$

(+2 pts)

c) Write the expression for the heat flux, q_z, in the slab. (2 pts)

Pulling directly from above: $\boxed{q_z = \frac{T_2 - T_1}{H}}$

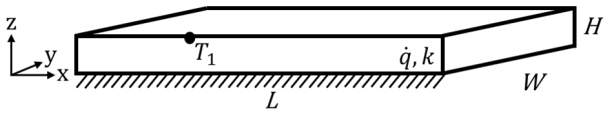
(+2 pts)

d) Sketch the temperature profile in the slab. (2 pts)

Linear temperature profile between $T(z=0) = T_1$ and $T(z=H) = T_2$
(+2 correct profile)

Problem 2 (30 pts)

a) A thin polysilicon resistive film of thickness H ($L, W >> H$) is bonded to an insulating substrate. Current is passed through the polysilicon film, causing uniform heat generation in the slab, $\dot{q}$. The top surface is maintained at T_1.



i) In which direction is heat transported? What is causing the heat transport? (2 pts)

Heat is transported from the insulated surface outwards. Conduction once more is causing this heat transport.
(+1 heat transported up away from insulated surface, +1 conduction via microscopic collisions between molecules with non-equal thermal energies)

ii) Starting from the generalized energy balance, derive the steady state temperature profile, $T(z)$, in the slab. (4 pts)

BCs:
$T(z=H) = T_1$ (+1 pt), $q_z|_{z=0} = 0$, or $\frac{dT}{dz}|_{z=0} = 0$ (+1 pt)
Shell balance method:
in-out+generation-consumption=accumulation
in + generation = out
$q_z|_z \cdot LW + \dot{q} \cdot LW\Delta z = q_z|_{z+\Delta z} \cdot LW$
$\frac{q_z|_z}{\Delta z} + \dot{q} = \frac{q_z|_{z+\Delta z}}{\Delta z}$
$\dot{q} = \frac{q_z|_{z+\Delta z} - q_z|_z}{\Delta z}$ $\lim_{\Delta z \to 0} \frac{q_z|_{z+\Delta z} - q_z|_z}{\Delta z} = \frac{dq_z}{dz} = \dot{q}$
$\dot{q} = -k\frac{d^2T}{dz^2} \rightarrow \int \dot{q} dz = -k \int d(\frac{dT}{dz}) \rightarrow \frac{dT}{dz} = -\frac{\dot{q}}{k} z + C_1$
Apply no flux BC: $0 = -\frac{\dot{q}}{k}(0) + C_1 \rightarrow C_1 = 0$. Integrate $\frac{dT}{dz} = -\frac{\dot{q}}{k} z$ again,
$T = -\frac{\dot{q}}{2k} z^2 + C_2$, apply fixed temperature BC: $T_1 = -\frac{\dot{q}}{2k} H^2 + C_2$
$C_2 = T_1 + \frac{\dot{q}}{2k} H^2 \rightarrow \boxed{T(z) = \frac{\dot{q}}{2k}(H^2 - z^2) + T_1}$ (+1 for method, +1 for answer)
Governing equation method:
$\rho \hat{c}_p \frac{\partial T}{\partial t} = k(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}) + \dot{q}$
$-\dot{q} = k\frac{\partial^2 T}{\partial z^2}$ (+1 pt)

Follow same integrating method as above: $\boxed{T(z) = \frac{\dot{q}}{2k}(H^2 - z^2) + T_1}$ (+1 for answer)

iii) Sketch the temperature profile in the slab. (2 pts)

Parabolic with maximum at insulated face (+ derivative=0), minimum at the outside boundary.
(+1 Pt for maximum at insulated face, +1 pt for parabolic with $\frac{dT}{dz} = 0$ at insulated face.)

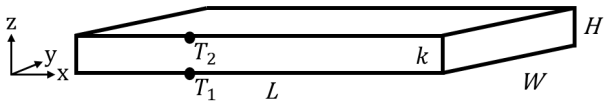
iv) Where is the temperature the highest? Why is the temperature highest there? (2 pts)

The temperature is the highest closest to the insulated face. This makes sense because there are fewer directions the heat can escape compared to the rest of the volume. (+2 pts)

v) How would the shape of the temperature profile change if the rate of heat generation were higher (i.e. double $\dot{q}$)? What about if the rate of heat generation were of the form $\dot{q} = Az$? (2 pts)

If the rate of heat generation is increased, the curvature of the temperature profile would increase and the maximum temperature reached at the insulated wall would increase. In this case, the temperature profile would be cubic. (+2 pts)

b) The same current-passing film $(H, L, W, k, \dot{q})$ is cast elsewhere in the device such that the upper and lower surfaces are held at T_2.



i) Sketch the shape of the temperature profile in the slab. In which direction is heat being transported and what is causing the heat transport? (4 pts)

Parabolic (concave down), centered about $z = H/2$. Heat is transported from the center of the slab outward, again conduction is causing the heat transport.
(+2 Correct Profile, +2 Heat transported outward from the middle via conduction)

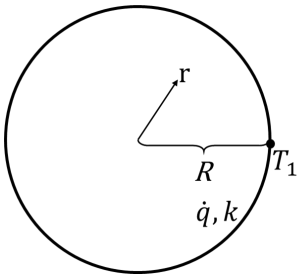
ii) Starting from the generalized energy balance, solve for the temperature profile in the film. (4 pts)

Correct BCs: $T(z = 0) = T_2, T(z = H) = T_1$ (+2 pt)
Same shell balance or governing EQ simplification as before, yielding $\dot{q} = -k\frac{d^2T}{dz^2}$. (+1 pt).
Integrate to get same expression as before, $\frac{dT}{dz} = -\frac{\dot{q}}{k}z + C_1$.
Integrate again: $T = -\frac{\dot{q}}{2k}z^2 + C_1 z + C_2$. Apply BCs $\rightarrow T_2 = C_2$
$T = -\frac{\dot{q}}{2k}z^2 + C_1 z + T_2 \rightarrow\ T_1 = -\frac{\dot{q}}{2k}H^2 + C_1 H + T_2, \rightarrow C_1 = \frac{T_1 - T_2}{H} + \frac{\dot{q}}{2k}H$
$\boxed{T(z) = \frac{\dot{q}}{2k}(Hz - z^2) + T_2 + \frac{T_1 - T_2}{H}z.}$ (+2 for process, +2 for answer)

iii) What is the heat flux at the center of the slab? Why? (2 pts)

The heat flux is 0 at the center of the slab. This is due to the symmetry in the temperature profile. At the center, the direction of heat transport flips.
(+2 Flux is Zero)

c) Another device uses a spherical solder bump of radius R and thermal conductivity k to pass current between two substrates, generating heat uniformly throughout its volume at a rate $\dot{q}$. Its external surface is held at a constant temperature, T_1.



i) Starting from the generalized energy balance, derive the steady state profile, $T(r)$, in the sphere. (4 pts)

BCs:
$T(r=R)=T_1$ (+1 pt), $\frac{dT}{dr}|_{r=0}=q_r=0$ (no heat flux in middle of sphere) (+1 pt).
Shell balance method: in-out+generation-consumption=accumulation
in+generation=out
$q_r|_r \cdot 4\pi r^2 + \dot{q}\cdot 4\pi r^2\cdot\Delta r = q_r|_{r+\Delta r}\cdot 4\pi(r+\Delta r)^2$ (assuming $dV\approx 4\pi r^2\Delta r$ in limit of $\Delta r\to 0$).
$\dot{q}=\frac{q_r|_{r+\Delta r}(r+\Delta r)^2/r^2 - q_r|_r}{\Delta r}$, $\lim_{\Delta r\to 0}\frac{q_r|_{r+\Delta r}(r+\Delta r)^2/r^2 - q_r|_r}{\Delta r} = \lim_{\Delta r\to 0}\frac{1}{r^2}\frac{q_r|_{r+\Delta r}(r+\Delta r)^2 - q_r|_r r^2}{\Delta r} =$
$\frac{1}{r^2}\frac{d}{dr}(r^2 q_r)=\dot{q}$ (+2 pts)
Separate and integrate: $\int \dot{q}r^2 dr = \int d(r^2 q_r) \to \frac{\dot{q}}{3}r^3 + C_1 = r^2 q_r = -kr^2\frac{dT}{dr}$.
$-\frac{\dot{q}}{3k}r + \frac{C_1}{r^2} = \frac{dT}{dr}$. BC 1: no flux at center (finite heat flux) $\to C_1 = 0$.
Integrate again, $\to T = -\frac{\dot{q}}{6k}r^2 + C_2$.
BC 2: $T(r=R)=T_1 \to T_1 = -\frac{\dot{q}}{6k}R^2 + C_2$, $C_2 = T_1 + \frac{\dot{q}}{6k}R^2$.
$\boxed{T(r)=\frac{\dot{q}}{6k}(R^2-r^2)+T_1}$ (+1 for method, +1 for answer)
General balance method: $\rho\hat{c}_p\frac{\partial T}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}(kr^2\frac{\partial T}{\partial r}) + \frac{1}{r^2 sin\theta}\frac{\partial}{\partial\theta}(ksin\theta\frac{\partial T}{\partial\theta}) + \frac{1}{r^2 sin^2\theta}\frac{\partial}{\partial\phi}(k\frac{\partial T}{\partial\phi}) + \dot{q}$
Assuming steady state, spherically symmetric, constant k:
$0 = k\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial T}{\partial r}) + \dot{q} \to -\frac{\dot{q}}{k}r^2 = \frac{d}{dr}(r^2\frac{dT}{dr})$. Integrate: $-\frac{\dot{q}}{3k}r^3 + C_1 = r^2\frac{dT}{dr}$ (same integration as shell method from here) $\to \boxed{T(r)=\frac{\dot{q}}{6k}(R^2-r^2)+T_1}$ (+1 for method, +1 for answer)

ii) Where is the temperature highest in the sphere? (2 pts)

The sphere is hottest in the center. There is the least freedom for heat to conduct away from the center. (+2 pts)

iii) Based on your analysis from above, draw the temperature profile in the sphere with a solid line. With a dotted line, draw the profile if there were no heat generation. (2 pts)

With Heat Generation: Concave down and parabolic with minimum at the exterior and maximum at the center.
(+1 point correct)

No Heat Generation: Flat Line at T_1. With no heat source the sphere will equilibrate at the temperature set by the boundary condition at $r=R$.
(+1 point correct)

Problem 3 (15 pts)

Helium is a critical as an inert gas for making semiconductors and for cooling the superconducting magnets that power MRI machines and NMR spectrometers. In nature, helium is produced through radioactive decay of Uranium and Thorium, which subsequently appears as a dilute component in natural gas. The largest Helium production facility in the United States is the Schute Creek Helium Plant in Wyoming. Schute Creek produces 1.4 billion cubic feet of liquid helium every year, making up 20 percent of global supply. The natural gas mixture at Schute Creek is unique, with CO_2 as the primary component rather than Methane (CH_4). Using the supplied data about Schute Creek's natural gas composition, answer the following questions.

Table 1: Natural Gas Mixture at the Schute Creek Helium Plant

Component	Mole Percent
CO_2	65%
CH_4	21.4%
N_2	7%
H_2S	5%
He	0.6%

a) Calculate the mass fraction of each component in the mixture. (3 pts)

Calculating mass fraction (ω_i) from component mass (m_i) and total mass (m_{tot})

$$\omega_i = \frac{m_i}{m_{tot}}$$

Calculating component mass from molar composition (n_i) and molar mass (MW_i)

$$m_i = n_i MW_i$$

Calculations for the Schute creek Gas Mixture using a basis of 1 total mole

$$m_{CO_2} = n_{CO_2} MW_{CO_2} = (0.65\ mol)(44.0\ \tfrac{g}{mol}) = 28.6\ g$$
$$m_{CH_4} = n_{CH_4} MW_{CH_4} = (0.214\ mol)(16.04\ \tfrac{g}{mol}) = 3.43\ g$$
$$m_{N_2} = n_{N_2} MW_{N_2} = (0.07\ mol)(28.0\ \tfrac{g}{mol}) = 1.96\ g$$
$$m_{H_2S} = n_{H_2S} MW_{H_2S} = (0.05\ mol)(34.08\ \tfrac{g}{mol}) = 1.70\ g$$
$$m_{He} = n_{He} MW_{He} = (0.006\ mol)(4.00\ \tfrac{g}{mol}) = 0.024\ g$$

(+0.5 pts per ansewr, +0.5 for method)

b) Determine the average molar mass of the mixture. (3 pts)

Calculating Average Molar Mass ($\overline{m}$) from total mass (m_{tot}) and total number of moles (n_{tot})

$$m_{tot} = \sum_i^n m_i = 35.7\ g$$
$$\overline{m} = \frac{m_{tot}}{n_{tot}} = \frac{35.7\ g}{1\ mol} = 35.7 \frac{g}{mol}$$

(+2 Points Answer, +1 Point Formula)

c) As this gas escapes from underground, it has a wellhead pressure of 2000 psi and a temperature of 275 $^\circ F$. Calculate the density of the escaping ($\rho_{wellhead}$) gas, and concentration of Helium within that gas ($C_{He,wellhead}$). (5 pts)

Ideal gas law: $PV = nRT$

Rearranging the ideal gas law to solve for concentration: $\frac{n}{V} = C = \frac{P}{RT}$

Converting Pressure and Temperature to appropriate units

$$P_{wellhead} = 2000\ psi\ \times \frac{1\ atm}{14.7\ psi} = 136\ atm$$
$$T_{wellhead} = 275^\circ F = 408\ K$$

Calculating Total Gas Concentration

$$C_{wellhead} = \frac{P_{wellhead}}{RT_{wellhead}} = \frac{136\ atm}{(8.206\times10^{-5}\ \frac{m^3\ atm}{K\ mol})(408\ K)} = 4060\ \frac{mol}{m^3}$$

Calculating Total Gas Density

$$\rho_{wellhead} = C_{wellhead} \times \overline{m} = 4060\ \frac{mol}{m^3} \times 35.7 \frac{g}{mol} \times \frac{1\ kg}{1000\ g} = \boxed{145 \frac{kg}{m^3}}$$

Calculating Helium Concentration

$$C_{He,wellhead} = C_{wellhead} \times y_{He} = 4060 \frac{mol}{m^3} \times 0.06 = \boxed{244 \frac{mol}{m^3}}$$

(+2 pts for $\rho_{wellhead}$, +2 pts for $C_{He,wellhead}$, +1 pt for correct approach)

d) As natural gas escapes the ground, it experiences an isothermal drop in pressure. What are the directional changes (increase, decrease, no change) you expect to see for the gas density, average molar mass, and mass-averaged velocity. (4 pts)

The table below describes the directional changes expected for the gas experienceing a decrease in pressure.

Property	Directional Change	Justification
Density (ρ)	$\downarrow$	Per the Ideal Gas Law, as pressure decreases the molar concentration and thus density will decrease
Average Molar Mass ($\overline{m}$)	No Change	Average molar mass is not a function of pressure
Mass-Averaged Velocity (v)	No Change	Mass-Averaged velocity represents the net flow of mass within a mixture. The rate of mass flow doesn't change as a function of pressure

(+0.5 pts per correct direction, +0.5 per correct justification, +1 pt for complete)

Problem 4 (20 pts)

The Diffusion of large solutes at infinite dilution can be described using the Stokes-Einstein equation. In the human body, Hemoglobin is a critical oxygen-transporting protein in blood. The diffusion constant $D_{heme,exp}$ for hemoglobin in water at body temperature ($T_{body} = 98.6°F$) is $6.9 * 10^{-11} \frac{m^2}{s}$. You may assume that the viscosity of water at T_{body} is 0.692 cP.

a) Using Stokes-Einstein, calculate the hydrodynamic diameter of hemoglobin ($d_{heme,est}$) at T_{body}. (8 pts)

Recall the Stokes-Einstein equation: $D_{A,B} = \frac{k_B T}{6\pi r_A \mu_B}$

Rearrange to solve for solute (A) radius: $r_A \mu_B = \frac{k_B T}{6\pi D_{A,B} \mu_B}$

Identify relevant constants and convert to useful units

$\mu_B = 0.692\ cP = 6.92 \times 10^{-4} \frac{Ns}{m^2}$; $k_B = 1.381 \times 10^{-23} \frac{J}{K}$; $T_{body} = 310\ K$

Calculate $d_{heme,est}$ using the re-arranged Stokes-Einstein equation

$$d_{heme,est} = 2r_{heme,est} = \frac{(1.381\times10^{-23}\frac{J}{K})(310\ K)}{3\pi(6.9*10^{-11}\frac{m^2}{s})(6.92\times10^{-4}\frac{Ns}{m^2})} = \boxed{9.51\ nm}$$

(+3 Correct Stokes-Einstein, +3 Correct $d_{heme,est}$, +2 for Complete)

b) Researchers used Dynamic Light Scattering (DLS) methods to experimentally measure the hydrodynamic diameter of Hemoglobin ($d_{heme,exp} = 5.5\ nm$). Based on this experimentally measured diameter, use Stokes-Einstein to calculate an estimated diffusion constant $D_{heme,est}$ for hemoglobin at T_{body}. (5 pts)

Apply Stokes-Einstein to calculate$D_{heme,est}$

$$D_{heme,est} = \frac{k_B T}{6\pi \frac{d_{heme,exp}}{2} \mu_B} = \frac{(1.381\times10^{-23}\frac{J}{K})(310\ K)}{3\pi(5.5\times10^{-9}\ m)(6.92\times10^{-4}\frac{Ns}{m^2})} = \boxed{1.19 \times 10^{-10} \frac{m^2}{s}}$$

(+2 pts Correct Answer,+2 pts Correct Stokes-Einstein, +1 pt Correct Units)

c) List and justify two reasons that the estimated diffusion constant for hemoglobin $D_{heme,est}$ differs from the experimentally measured value $D_{heme,exp}$ (7 pts)

$D_{heme,exp}$ is only 57 percent of $Dheme, est$. This means that hemoglobin moves slower in water than the Stokes-Einstein equation predicts. There are many reasons this difference may arise, primarily due to the assumptions used to develop the Stokes-Einstein equation.

(+3 points for recognizing that hemoglobin moves slower than predicted)

Two example justifications include:

Intermolecular Interactions: Stokes-Einstein assumes the only interactions between solute and solvent are viscous forces. However, water can form strong hydrogen bonds with Hemoglobin, adding additional forces which can have the effect of slowing Diffusion.

Non-Sphericity of Hemoglobin: Stokes-Einstein assumes solutes are perfect spheres. Hemoglobin, which takes the shape of an oblong donut, isn't a perfect sphere and thus will have diffusive behavior which differs from the Stokes-Einstein assumptions.

(+2 pts per worth assumption)

Problem 5 (25 pts)

Determine the value of the following gas diffusion constants using the Hirschfelder Equation.

Recall the Hirschfelder Equation:

$$\boxed{D_{A,B} = \frac{0.001858 T^{3/2} (\frac{1}{M_A} + \frac{1}{M_B})^{1/2}}{P \sigma_{AB}^2 \Omega_D}}$$

Use the equations for $\sigma_A B$ and ϵ_{AB}, along with the figure for determining Ω_D provided in the diffusivity handout.

a) Ar in Air on the top of Mt. Everest (250 torr) at 0 $°C$ (7 pts)

Relevant Physical Properties (Appendix K):

Argon: $\sigma_{Ar} = 3.418\ \mathring{A}$, $\frac{\epsilon_{Ar}}{k} = 124\ K$, $M_{Ar} = 39.948 \frac{g}{mol}$

Air: $\sigma_{Air} = 3.617\ \mathring{A}$, $\frac{\epsilon_{Air}}{k} = 97\ K$, $M_{Air} = 28.965 \frac{g}{mol}$

Calculated Hirschfeld Parameters

$\sigma_{Ar,Air} = 3.518\ \mathring{A}$, $\frac{\epsilon_{Ar,Air}}{k} = 109.7\ K$, $\boxed{\Omega_{D,Air,Ar}(T = 273\ K) = 1.00}$, $P = 250\ torr \times \frac{1\ atm}{760\ torr} = 0.329\ atm$, $T = 0\ °C = 273\ K$

Hirschfeld Diffusivity Calculation

$$D_{Ar,Air} = \frac{0.001858(273\ K)^{3/2}(\frac{1}{39.948} + \frac{1}{28.965})^{1/2}}{(0.329\ atm)(3.518\mathring{A})^2(1.00)} = \boxed{0.502\ \frac{cm^2}{s}}$$

(+1 Uses Hirschfeld, +1 Correct Omega, +2 Correct Answer, +2 Correct Approach)

b) CO_2 in N_2 within a diving bell submerged $100 ft$ below the ocean's surface at 30 $°C$ (7 pts)

Calculating Hydrostatic Pressure 100 ft below the ocean's surface

Hydrostatic Equation: $P(h) = P_0 + \rho g h$

$P_0 = 1\ atm$, $g = 9.81 \frac{m^2}{s}$, $h = 100\ ft \times \frac{1\ m}{3.28\ ft} = 30.49\ m$

$\rho g h = (1000\ \frac{kg}{m^3})(9.81\ \frac{m^2}{s})(30.49\ m) = 299085\ Pa\ = 2.95\ atm$

$\boxed{P(100\ ft) = 1.00 + 2.95 = 3.95\ atm}$

Relevant Physical Properties (Appendix K):

CO_2: $\sigma_{CO_2} = 3.996\ \mathring{A}$, $\frac{\epsilon_{CO_2}}{k} = 190\ K$, $M_{CO_2} = 44.01 \frac{g}{mol}$

N_2: $\sigma_{N_2} = 3.681\ \mathring{A}$, $\frac{\epsilon_{N_2}}{k} = 91.5\ K$, $M_{N_2} = 28.013 \frac{g}{mol}$

Calculated Hirschfeld Parameters

$\sigma_{CO_2,N_2} = 3.839\ \mathring{A}$, $\frac{\epsilon_{CO_2,N_2}}{k} = 131.9\ K$, $\boxed{\Omega_{D,CO_2,N_2}(T = 303\ K) = 1.026}$, $T = 30\ °C = 303\ K$

Hirschfeld Diffusivity Calculation

$$D_{CO_2,N_2} = \frac{0.001858(303\ K)^{3/2}(\frac{1}{44.01} + \frac{1}{28.013})^{1/2}}{(3.95\ atm)(3.839\mathring{A})^2(1.026)} = \boxed{2.48 \times 10^{-2}\ \frac{cm^2}{s}}$$

(+1 Used Hirschfeld, +2 Correct Pressure, +1 Correct Omega, +1 Correct Answer, +2 Correct Approach)

c) SiH_4 in H_2 at 300 $°C$ and 300 mtorr. (7 pts)

Relevant Physical Properties (Appendix K):

SiH_4: $\sigma_{SiH_4} = 4.08\ \mathring{A}$, $\frac{\epsilon_{SiH_4}}{k} = 207.6\ K$, $M_{SiH_4} = 32.117 \frac{g}{mol}$

H_2: $\sigma_{H_2} = 2.968\ \mathring{A}$, $\frac{\epsilon_{H_2}}{k} = 33.3\ K$, $M_{H_2} = 2.02 \frac{g}{mol}$

Calculated Hirschfeld Parameters

$\sigma_{SiH_4,H_2} = 3.524\ \mathring{A}$, $\frac{\epsilon_{SiH_4,H_2}}{k} = 83.14\ K$, $\Omega_{D,SiH_4,H_2}(T = 573\ K) = 0.897$,
$P = 0.3\ torr \times \frac{1\ atm}{760\ torr} = 3.95 \times 10^{-4}\ atm$, $T = 300\ °C = 573\ K$

Hirschfeld Diffusivity Calculation

$$D_{SiH_4,H_2} = \frac{0.001858(573\ K)^{3/2}(\frac{1}{32.117} + \frac{1}{2.02})^{1/2}}{(3.95\times10^{-4}\ atm)(3.524\mathring{A})^2(0.897)} = \boxed{4161\frac{cm^2}{s}}$$

(+1 Used Hirschfeld, +1 Correct σ, +1 Correct Ω, +1 Correct Final Answer, +2 Correct Approach, +1 Complete)

d) Next summer you are an hired as an intern for a semiconductor manufacturer. Your team places you in charge of a deposition unit, which grows layers of Silicon using a SiH_4-H_2 gas mixture. You currently use the conditions outlines in part (c). If you drop operating pressure to $200 Torr$ and decrease temperature to 200 $°C$, how will the diffusion constant of SiH_4 to change? Use Hirschfeld extrapolation assuming no change to Ω_D. (4 pts)

Hirschfeld Extrapolation Equation

$$D_{A,B}(P_2, T_2) = D_{A,B}(P_1, T_1)(\frac{P_1}{P_2})(\frac{T_2}{T_1})^{3/2}$$

Applying Hirschfeld Extrapolation

$$D_{A,B}(200\ torr, 200\ °C) = 4161\frac{cm^2}{s}(\frac{300\ torr}{200\ torr})(\frac{473\ K}{573\ K})^{3/2} = \boxed{4681\ \frac{cm^2}{s}}$$

You expect the diffusion constant to increase.

(+1.5 Correct Use of Hirscheld Extrapolation Equation, +1.5 Correct Answer, +1, Correct Units)